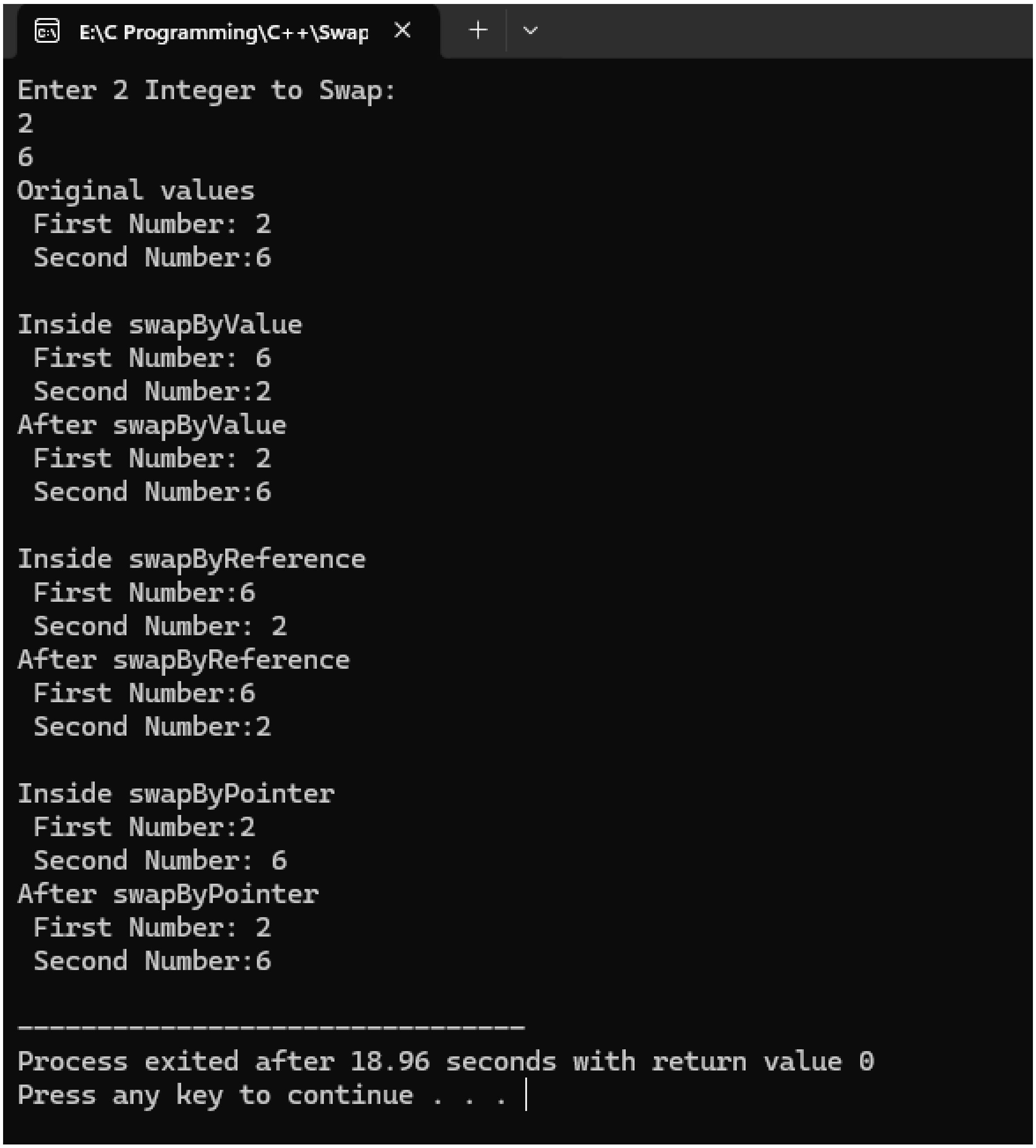
LAB-3

17). <u>// C++ program to Swap Two Numbers Using Concept of</u>
<u>//i). Pass By value</u>
<u>//ii). Pass By Refrence</u>
<u>//iii). Pass By Pointer</u>

```
#include <iostream>
using namespace std;

// i) Pass by Value
void swapByValue(int a, int b) {
   int temp = a;
   a = b;
   b = temp;
   cout << "Inside swapByValue \n First Number: " << a << "\n Second Number:" << b << endl;
}

// ii) Pass by Reference
void swapByReference(int &a, int &b) {
   int temp = a;
   a = b;
   b = temp;
   cout << "Inside swapByReference \n First Number:" << a << "\n Second Number: " << b << endl;
}

// iii) Pass by Pointer
void swapByPointer(int *a, int *b) {
   int temp = *a;
   *a = *b;
   *b = temp;
   cout << "Inside swapByPointer \n First Number:" << *a << "\n Second Number: " << *b << endl;
}

int main () {
   int x,y;
   cout<<"Enter 2 Integer to Swap:"<<endl;
   cin>>x>>y;
   cout << "Original values \n First Number: " << x << "\n Second Number:" << y <<"\n"<< endl;

   // Pass by Value
   swapByValue(x, y);
   cout << "After swapByValue \n First Number: " << x << "\n Second Number:" << y << endl << endl;

   // Pass by Reference
   swapByReference(x, y);
   cout << "After swapByReference \n First Number:" << x << "\n Second Number:" << y << endl <<
endl;

   // Pass by Pointer
```

```
    swapByPointer(&x, &y);
    cout << "After swapByPointer \n First Number: " << x << "\n Second Number:" << y << endl;

    return 0;
}
```

Output:



LAB-3

18). // C++ program to Find Simple Interest Using Concept of
//i). Return By value
//ii). Return By Refrence
//iii). Return By Pointer

```
#include <iostream>
using namespace std;

// i) Return by Value
float Value(float p, float r, float t) {
   float si = (p * r * t) / 100;
   return si;
}

// ii) Return by Reference
float siRef;
float& Reference(float p, float r, float t) {
   siRef = (p * r * t) / 100;
   return siRef;
}

// iii) Return by Pointer (using static variable)
float* Pointer(float p, float r, float t) {
   static float si;
   si = (p * r * t) / 100;
   return &si;
}

int main () {
   float p, r, t;
   cout << "Enter the Principal, Rate and Time (in years): ";
   cin >> p >> r >> t;
   cout << "\nPrincipal: " << p << ", Rate: " << r << ", Time: " << t << endl;

   float valueResult = Value(p, r, t);
   cout << "Simple Interest (Return by Value): " << valueResult << endl;

   float& referenceResult = Reference(p, r, t);
   cout << "Simple Interest (Return by Reference): " << referenceResult << endl;

   float* pointerResult = Pointer(p, r, t);
   cout << "Simple Interest (Return by Pointer): " << *pointerResult << endl;
```

```
    return 0;
}
```

- Output:

```
E:\C Programming\C++\Simp

Enter the Principal, Rate and Time (in years): 123999.00
12.0
5.0

Principal: 123999, Rate: 12, Time: 5
Simple Interest (Return by Value): 74399.4
Simple Interest (Return by Reference): 74399.4
Simple Interest (Return by Pointer): 74399.4

--------------------------------
Process exited after 19.54 seconds with return value 0
Press any key to continue . . .
```

LAB-3

19). // C++ Simple Program Showing Implicit and Explicit Types conversion

```
#include<iostream>
using namespace std;

int main(){
            int a=400;
            double b= 987654321.234;
// Implict Type Conversion
  int c=b;
  double d=a;
            cout<<"The Implicit Conversion (double to int):" <<c<<endl;
            cout<<"The Implicit Conversion (int to double ):" <<d<<endl;
//Explicit Type Conversion
  int x =(int)b;
  double y =double (a);
            cout<<"The Explicit Conversion (double to int):" <<x<<endl;
            cout<<"The Explicit Conversion (int to double):" <<y<<endl;
            return 0;
}
```

✶ Output: